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The sharp A_p bound for the Hardy–Littlewood maximal operator

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Weights and weighted L^p spaces

Definition 1 (Weights)

A *weight* is a function $w \in L^1_{\text{loc}}(\mathbb{R}^d)$ with $w \geq 0$ that is not almost everywhere zero.

Definition 2 (Weighted L^p spaces)

Let w be a weight and let $1 \leq p < \infty$. The weighted L^p space is

$$L^p_w(\mathbb{R}^d) := \left\{ f : \mathbb{R}^d \to \mathbb{R} \text{ measurable} : \int_{\mathbb{R}^d} |f(y)|^p \, w(y) \, dy < \infty \right\},$$

with norm $\|f\|_{L^p_w} := \left(\int_{\mathbb{R}^d} |f(y)|^p \, w(y) \, dy\right)^{1/p}$. Moreover, $L^\infty_w(\mathbb{R}^d)$ is the space of measurable functions f with $\|f\|_{L^\infty_w} := \inf\{t \geq 0 : w(\{|f| > t\}) = 0\} < \infty$.

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The Hardy–Littlewood maximal operator

Definition 3 (Hardy–Littlewood maximal operator)

$$M_w f(x) := \sup_{Q \ni x} \frac{1}{w(Q)} \int_Q |f(y)|\, w(y)\, dy.$$

M_w^c takes the supremum over centred cubes $Q(x, r)$ only, and $w \equiv 1$ gives M and M^c.

M bounded on $L^p_w(\mathbb{R}^d)$

$$\|Mf\|_{L^p_w} = \left(\int_{\mathbb{R}^d} (Mf)^p \, w \, dy \right)^{1/p} \leq C \left(\int_{\mathbb{R}^d} |f|^p \, w \, dy \right)^{1/p} = C \, \|f\|_{L^p_w}$$

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A necessary condition on w

Proposition 1 (Muckenhoupt)

If M is bounded on $L^p_w(\mathbb{R}^d)$ with constant C, then every cube Q satisfies

$$\left(\frac{1}{|Q|}\int_Q w\,dy\right)\left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\,dy\right)^{p-1} \leq C^p.$$

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Definitions 4 and 5 (A_p characteristic, dual weight)

$$[w]_{A_p} := \sup_Q \left(\frac{1}{|Q|}\int_Q w\,dy\right)\left(\frac{1}{|Q|}\int_Q w^{-\frac{1}{p-1}}\,dy\right)^{p-1} = \sup_Q \frac{w(Q)}{|Q|}\left(\frac{\sigma(Q)}{|Q|}\right)^{p-1},$$

where $\sigma := w^{-1/(p-1)}$ is the *dual weight* of w, and $w \in A_p$ if $[w]_{A_p} < \infty$.

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A_p is necessary and sufficient

Corollary 1

If M is bounded on $L^p_w(\mathbb{R}^d)$ with constant C, then $w \in A_p$ with

$$[w]_{A_p} \leq C^p \quad \Longleftrightarrow \quad [w]_{A_p}^{\frac{1}{p}} \leq C.$$

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Theorem 1 (Muckenhoupt 1972)

Let $1 < p < \infty$ and w a weight. Then M is bounded on $L^p_w(\mathbb{R}^d)$ iff $w \in A_p$.

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Main result

Theorem 2 (Buckley 1993, Lerner 2008)

Let $1 < p < \infty$.

(i) *There is a constant* $D_{p,d}$*, depending only on* p *and* d*, such that for every* $w \in A_p$ *and every* $f \in L^p_w(\mathbb{R}^d)$*,*

$$\|Mf\|_{L^p_w} \leq D_{p,d}\,[w]_{A_p}^{\frac{1}{p-1}}\,\|f\|_{L^p_w}.$$

(ii) *If* $\alpha < \frac{1}{p-1}$*, then there is no constant* D *such that* $\|Mf\|_{L^p_w} \leq D\,[w]_{A_p}^{\alpha}\,\|f\|_{L^p_w}$ *for all* $w \in A_p$ *and all* $f \in L^p_w(\mathbb{R}^d)$.

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